Thermal Expansion Quick Reference for Piping and Brackets

Document No: GDE-THM-01 Revision: A Date: 2022-11-20

Linear expansion formula

Change in length: delta L = alpha x L x delta T, where alpha is the coefficient of linear thermal expansion (1/K), L is the original length (m) and delta T is the temperature change (K).

Typical coefficients (20 to 100 deg C)

Material	alpha (1/K)
Carbon steel	12.0e-6
Stainless steel 304	17.3e-6
Aluminium 6061	23.6e-6
Silicon carbide	4.0e-6

Worked example

A 2 m aluminium 6061 bracket heated from 20 deg C to 120 deg C expands by 23.6e-6 x 2 x 100 = 4.72 mm if unrestrained. If restrained, thermal stress sigma = E x alpha x delta T.